

A Level Mechanics

Notes · Edexcel A Level Mathematics — Mechanics · Chapter 3: Forces and Newton's Laws of Motion

Chapter 3: Forces and Newton's Laws of Motion

Why this chapter matters

Forces are the cause of every acceleration studied in mechanics, and Newton's three laws of motion are the tools that connect the forces acting on a body to how it moves. This chapter builds three core skills: drawing a clear force diagram, resolving forces along a chosen direction, and applying $F = ma$ to a single particle, to two particles connected by a string, and to particles connected over a pulley. Almost every later chapter — moments, friction on inclined planes, projectiles — starts from a correctly drawn force diagram and a correctly applied Newton's second law equation, so it is worth mastering the habits in this chapter before moving on.

3.1 Types of force and force diagrams

Concept

A force diagram (also called a free-body diagram) shows every force acting on a single particle as an arrow drawn from a point representing that particle. Each arrow is labelled with a name or symbol and points in the direction the force acts. Only forces acting **on** the particle itself are shown — never a force the particle exerts on something else (that pairing is Newton's third law, covered in section 3.3).

The standard forces met at A Level are:

- **Weight (W)** — the force of gravity, always vertically downward, $W = mg$.
- **Normal reaction (R or N)** — the contact force a surface exerts on a particle, always perpendicular to that surface, pushing away from it.
- **Tension (T)** — the pulling force in a taut string or rope, directed along the string, away from the particle and towards the point of attachment.
- **Thrust** — the pushing force in a rigid rod or strut under compression, directed along the rod.
- **Friction (F)** — a contact force parallel to the surface, opposing relative motion or the tendency to slide.
- **Driving force** — a forward force produced by an engine, motor or applied push (for example, in car or train problems).
- **Resistance** — a general term for a force opposing motion, such as air resistance, acting opposite to the direction of travel.

A correct force diagram is usually worth marks in its own right, and it is the essential first step before any equation can be written down — get the diagram wrong, and every later step inherits the error.

Key vocabulary

Term	Meaning
------	---------

Force diagram / free-body diagram	A diagram showing every force acting on a particle as an arrow from a single point
Weight (W)	The force of gravity on a body, $W = mg$, always vertically downward
Normal reaction (R or N)	The contact force from a surface, perpendicular to that surface
Tension (T)	The pulling force in a taut string, rod or rope, directed along it, away from the particle
Thrust	The pushing (compressive) force in a rigid rod, directed along the rod
Friction (F)	A contact force parallel to the surface, opposing relative sliding or its tendency
Driving force	A forward force produced by an engine or applied push, e.g. on a car or cyclist
Resistance	A general opposing force, e.g. air resistance, acting opposite to the direction of motion

Key formulas

KEY FORMULA

$$W = mg$$

where W is weight in newtons (N), m is mass in kilograms (kg), and g is the acceleration due to gravity (usually taken as 9.8 m s^{-2} at A Level unless stated otherwise).

Procedure

1. Sketch the particle as a single point (or a small box/circle) and identify every object in contact with it or connected to it (surface, string, rod, another particle).
2. For each contact or connection, draw one arrow from the particle showing the force it produces: weight always vertically down; normal reaction perpendicular to, and away from, the contact surface; tension along a string, pulling towards the string's other end; friction parallel to the surface, opposing motion or the tendency to move.
3. Label every arrow with its name or symbol (W , R , T , F , and so on), and write its value next to it if the value is given in the question.
4. Check the diagram: no force should be drawn twice, and no force the particle exerts on something else should appear — only forces acting on the particle itself belong in its diagram.

Worked example

EXAMPLE 3.1

Question. A block rests on a rough horizontal floor. A string attached to the block is pulled

upward at 25° above the horizontal, and the block experiences resistance to sliding. Draw and describe the force diagram for the block, naming every force.

Step 1 — identify the contact with the floor. The block touches the floor, so it experiences weight (W , vertically down) and normal reaction (R , vertically up, perpendicular to the floor). **B1**

Step 2 — identify the string. The string is under tension, so the block experiences tension (T), acting along the string at 25° above the horizontal, away from the block — that is, up and to the side towards which it is pulled. **B1**

Step 3 — identify friction. The block tends to slide along the floor, so it also experiences friction (F), horizontal, acting opposite to the direction the block moves or tends to move. **B1**

Answer: The force diagram shows four forces acting on the block: weight W vertically downward, normal reaction R vertically upward, tension T at 25° above the horizontal (directed away from the block, along the string), and friction F horizontal, opposing the block's motion.

Exam tip

EXAM TIP

Force-diagram questions are marked strictly: an arrow with no label, or a diagram that shows a force acting on the string or the floor rather than on the particle, typically scores zero even when the geometry is drawn correctly. Build the habit of writing the force's name or symbol next to every arrow, and of asking “does this force act on the object I am modelling?” before drawing it. Marks are often lost here not because of misunderstanding the physics, but because the diagram is drawn quickly and the labelling step is skipped — treat the diagram as a piece of working that earns marks in its own right, not just a sketch to orient yourself.

Common exam mistakes

- Drawing a force twice, or showing the reaction pair (the force the particle exerts on the surface/string) instead of the force acting on the particle.
- Missing friction entirely when a surface is described as “rough” rather than “smooth”.
- Drawing tension pointing away from the string, or thrust pointing the wrong way along a rod.
- Leaving arrows unlabelled, which loses marks even when the diagram's geometry is otherwise correct.

3.2 Newton's first and second laws

Concept

Newton's first law states that a particle remains at rest, or moves with constant velocity, unless a resultant (net) force acts on it. In other words, zero resultant force means zero acceleration — this is the condition for equilibrium.

Newton's second law extends this to the case where the resultant force is not zero: the resultant force acting on a particle equals its mass times its acceleration,

$$F = ma$$

with the acceleration always in the same direction as the resultant force.

Applying either law requires two steps done in order: first draw the force diagram (section 3.1), then resolve forces in the direction(s) of interest — usually along the direction of motion and perpendicular to it. **Resolving** in one dimension simply means adding forces algebraically once a positive direction has been chosen. Resolving in two dimensions, for a force acting at an angle, means splitting that force into perpendicular components first — this works because any force can be replaced, without changing its effect, by two components along two perpendicular axes. Writing $R(\rightarrow)$ or $R(\uparrow)$ is a way of stating which axis you are summing forces along, so that the examiner (and you) can check every force has been projected onto it correctly.

Key vocabulary

Term	Meaning
Resultant force	The single force equivalent to the vector sum of all forces acting on a particle
Equilibrium	The state of a particle with zero resultant force, so zero acceleration (constant velocity or at rest)
Resolving	Splitting a force into components along chosen perpendicular directions, or summing forces along one direction
Newton (N)	The SI unit of force; $1 \text{ N} = 1 \text{ kg m s}^{-2}$
$R(\rightarrow)$, $R(\uparrow)$	Shorthand meaning “resolving in this direction”, used to introduce an equation of motion

Key formulas

KEY FORMULA

$$F = ma$$

$$\Sigma F = 0 \quad (\text{equilibrium})$$

where F is the resultant force (N), m is mass (kg), and a is acceleration (m s^{-2}).

Procedure

1. Draw the force diagram (section 3.1) and choose a positive direction — usually the direction of the acceleration, or the direction of motion.
2. Resolve every force along that direction (and perpendicular to it, if needed), writing $R(\rightarrow)$ or $R(\uparrow)$ to state which direction is being resolved.
3. Write the resulting equation: the sum of the resolved forces equals ma if the particle accelerates, or equals zero if it is in equilibrium.
4. Solve for the unknown (acceleration, an unknown force, or the normal reaction) and check the answer's sign matches the direction assumed.

Worked example

EXAMPLE 3.2

Question. A particle of mass 5 kg rests on a smooth horizontal surface. A force of 20 N is applied to the particle at 30° above the horizontal. Find (a) the acceleration of the particle, (b) the normal reaction between the particle and the surface.

Step 1 (part a) — resolve horizontally. $R(\rightarrow)$: only the horizontal component of the 20 N force acts in this direction, since the surface is smooth (no friction). **M1**

$$20 \cos 30^\circ = 5a$$

Step 2 — evaluate $\cos 30^\circ$ and solve for a. $\cos 30^\circ = 0.8660$ (4 s.f.), so: **M1**

$$20 \times 0.8660 = 5a \implies 17.32 = 5a$$

$$a = 17.32 \div 5 = 3.46 \text{ m s}^{-2} \text{ (3 s.f.) } \textbf{A1}$$

Step 3 (part b) — resolve vertically. $R(\uparrow)$: the particle does not accelerate vertically, so the resultant vertical force is zero. **M1**

$$R + 20 \sin 30^\circ - 5(9.8) = 0$$

Step 4 — evaluate $\sin 30^\circ$ and solve for R. $\sin 30^\circ = 0.5$, so: **M1**

$$R + 20(0.5) - 49 = 0 \implies R + 10 - 49 = 0$$

$$R = 49 - 10 = 39 \text{ N } \textbf{A1}$$

Answer: (a) $a = 3.46 \text{ m s}^{-2}$ (3 s.f.), in the direction of the horizontal component of the applied force. (b) $R = 39 \text{ N}$.

Exam tip

EXAM TIP

Always write the resolving direction — $R(\rightarrow)$ or $R(\uparrow)$ — as its own line before the equation, even when it feels obvious. It costs nothing, and it is frequently the method mark examiners are looking for, separate from the mark for the correct substitution. If you write the equation straight away, without stating which direction it resolves in, you can lose this mark even when your final answer is correct — the examiner cannot tell whether the direction was chosen correctly, or by luck.

Common exam mistakes

- Forgetting that the normal reaction is reduced (or increased) when part of another force acts vertically, as in part (b) above.
- Resolving a force at the wrong angle (using $\sin$ where $\cos$ is needed, or vice versa) — always check which side of the angle is “adjacent” to the direction being resolved.
- Not stating the direction being resolved, which can cost the method mark even when the substitution is correct.
- Mixing up mass (kg, scalar) and weight (N, vector force) inside an $F = ma$ equation.

3.3 Newton's third law and connected particles

Concept

Newton's third law states that if body A exerts a force on body B, then body B exerts an equal and opposite force on body A. The two forces of an action–reaction pair share four properties: they are equal in magnitude, opposite in direction, act on **different** bodies, and are of the same type (for example, both contact forces, or both gravitational).

Connected particles problems apply this alongside Newton's second law to two (or more) particles joined by a string, typically moving in a straight line — for example, a car towing a trailer, or two blocks joined by a string on a horizontal surface. Provided the string is modelled as **light** (massless) and **inextensible** (does not stretch), two key facts follow:

- The tension is the same throughout the string.
- Both particles have exactly the same speed and the same magnitude of acceleration at every instant.

These two facts let you write each particle's equation of motion separately, or write the whole system's equation at once by treating both particles as a single combined mass. Combining them this way eliminates the internal tension, because it is an internal force: it pulls forward on one particle and backward on the other with equal magnitude, so the two contributions cancel when the particles are considered together.

Key vocabulary

Term	Meaning
Action–reaction pair	Two equal and opposite forces, one on each of two interacting bodies (Newton's third law)
Connected particles	Two or more particles joined by a string (or rod) so they move with the same acceleration
Tension throughout	For a light, inextensible string, the tension is the same at every point along it
Equation of motion	The $F = ma$ equation written for one particle, or for the whole connected system
Whole-system equation	The combined equation for all connected particles together, in which internal tensions cancel

Key formulas

KEY FORMULA

$$T_{AB} = -T_{BA} \quad (\text{Newton's third law: equal and opposite})$$

$$\text{Whole system: } \sum F_{\text{external}} = (m_A + m_B)a$$

Procedure

1. Confirm (or state as a modelling assumption) that the string is light and inextensible, so the tension is the same throughout and both particles share one acceleration, a .
2. Draw a separate force diagram for each particle, including the tension acting on each end of the string (equal in magnitude, opposite in direction on each particle, by Newton's third law).
3. Either write the whole-system equation (external forces only, combined mass) to find a directly, or write each particle's individual equation of motion and solve simultaneously.
4. Use the individual equation for whichever particle was not used to find a , to find the tension T ; check the answer by substituting into the other particle's equation.

Worked example

EXAMPLE 3.3

Question. Particles A (mass 3 kg) and B (mass 2 kg) are connected by a light, inextensible string and lie on a smooth horizontal surface. A horizontal force of 20 N is applied to A, pulling the system so that B follows A, connected by the string. Find the acceleration of the system and the tension in the string.

Step 1 — state the modelling assumption. The string is light and inextensible, so the tension T is the same throughout, and both particles have the same acceleration a . **B1**

Step 2 — write the whole-system equation. $R(\rightarrow)$, combining both particles (internal tension cancels, so it does not appear). **M1**

$$20 = (3 + 2)a$$

Step 3 — solve for a . **A1**

$$a = 20 \div 5 = 4 \text{ m s}^{-2}$$

Step 4 — write particle B's equation to find T . $R(\rightarrow)$ for B alone; only the tension acts on B in the horizontal direction. **M1**

$$T = 2a = 2(4) = 8 \text{ N} \quad \textbf{A1}$$

Step 5 — check using particle A's equation. **B1**

$$20 - T = 3a \implies 20 - 8 = 12 = 3(4) \quad \checkmark$$

Answer: $a = 4 \text{ m s}^{-2}$, $T = 8 \text{ N}$.

Exam tip

EXAM TIP

Always write the modelling assumption — “the string is light and inextensible, so T is the same throughout and both particles have the same acceleration” — as an explicit line of working before the equations, rather than treating it as obvious and skipping straight to $F = ma$. Examiners often award a separate mark for stating it, and writing it out first also gives you a checklist: it reminds you that tension is internal (so it must cancel in the whole-system equation) and that both particles' accelerations must be equal in magnitude before you start combining equations.

Common exam mistakes

- Forgetting that tension is an internal force and must not appear in the whole-system equation.
- Using the wrong sign for tension on one of the two particles (it must oppose the direction the string pulls each particle).
- Not checking the answer against the second particle's equation, which is a fast way to catch an arithmetic slip.
- Assuming the string stays taut (and tension is positive) without checking — if a calculated tension comes out negative, the string has gone slack and the model breaks down.

3.4 Pulleys

Concept

A **smooth, light, fixed pulley** changes the direction of a string's tension without changing its magnitude, and does no work itself. The tension is the same on both sides of the pulley, provided the pulley is smooth (frictionless) and light (massless), and the string is inextensible.

Two standard setups appear repeatedly at A Level:

- **Two particles hanging vertically** from a string passing over a pulley. The heavier particle accelerates downward and the lighter one accelerates upward, with the same magnitude of acceleration, since the string is inextensible.
- **One particle on a horizontal table**, connected by a string over a pulley at the table's edge to a second particle hanging freely. The hanging particle's weight drives the system, resisted only by the inertia of both masses if the table is smooth.

In every pulley problem, the method is the same as for connected particles (section 3.3): draw each particle's force diagram separately, define one consistent positive direction of motion (following the string, over the pulley, from one particle to the other), write each particle's equation of motion, and solve simultaneously.

Key vocabulary

Term	Meaning
Smooth pulley	A frictionless pulley — the tension is unchanged as the string passes over it
Light pulley	A massless pulley — no force is needed to accelerate the pulley itself
Fixed pulley	A pulley whose axle does not move (as opposed to a “moving” pulley, not required at this level)
Positive direction	The single direction, followed around the pulley, in which both particles' motion is taken as positive

Key formulas

KEY FORMULA

Two hanging masses ($m_1 > m_2$): $a = (m_1 - m_2)g \div (m_1 + m_2)$

$$T = 2m_1m_2g \div (m_1 + m_2)$$

Mass on smooth table (m_1) with hanging mass (m_2): $a = m_2g \div (m_1 + m_2)$

$$T = m_1m_2g \div (m_1 + m_2)$$

These shortcut formulas are useful for checking an answer quickly, but on their own they do not show the working an examiner is looking for — always derive them from the individual equations of motion first, as in the worked example below.

Procedure

1. Draw a separate force diagram for each particle, including its own weight and the tension (the same magnitude on each particle, since the pulley is smooth and light).
2. Define one positive direction of motion that runs consistently through the string and over the pulley (for example, “downward for A, which is the same as upward for B, is positive”).
3. Write each particle's equation of motion in that positive direction.
4. Solve the two equations simultaneously to find a , then substitute back into either equation to find T ; check with the other equation.

Worked example

EXAMPLE 3.4

Question. Particles A (mass 5 kg) and B (mass 3 kg) are connected by a light, inextensible string passing over a smooth fixed pulley, and hang freely on either side of the pulley. The system is released from rest. Find the acceleration of the system and the tension in the string. (Take $g = 9.8 \text{ m s}^{-2}$.)

Step 1 — decide the direction of motion. A is heavier, so A accelerates downward and B accelerates upward; take this as the positive direction for each particle. The pulley is smooth and light, so T is the same on both sides. **B1**

Step 2 — write particle A's equation. $R(\downarrow, \text{positive})$ for A. **M1**

$$5(9.8) - T = 5a \implies 49 - T = 5a$$

Step 3 — write particle B's equation. $R(\uparrow, \text{positive})$ for B. **M1**

$$T - 3(9.8) = 3a \implies T - 29.4 = 3a$$

Step 4 — add the two equations to eliminate T . Adding the left-hand sides and the right-hand sides of Steps 2 and 3: **M1**

$$(49 - T) + (T - 29.4) = 5a + 3a \implies 49 - 29.4 = 8a$$

$$19.6 = 8a \implies a = 19.6 \div 8 = 2.45 \text{ m s}^{-2} \quad \text{A1}$$

Step 5 — substitute back into particle B's equation to find T . **M1**

$$T = 29.4 + 3(2.45) = 29.4 + 7.35 = 36.75 \text{ N} \quad \text{A1}$$

Answer: $a = 2.45 \text{ m s}^{-2}$, $T = 36.75 \text{ N}$.

Exam tip

EXAM TIP

Write out both particles' individual equations of motion in full (Steps 2–3 above) before combining them, rather than jumping straight to the combined formula. Examiners give separate method marks for each particle's equation and a further mark for correctly combining them, so this order protects your method marks even if a later step contains an arithmetic slip — and it also gives you a second equation to check your final answers against.

Common exam mistakes

- Using inconsistent positive directions for the two particles (they must be consistent around the pulley — “up” for one side is “down” for the other).
- Quoting the shortcut formulas for a directly without showing the individual equations of motion, which loses method marks even if the final answer is correct.
- Forgetting that a mass resting on a table (rather than hanging) has no weight component along the string's direction on a smooth horizontal table — only the tension acts on it horizontally.
- Not checking whether the calculated tension is positive and less than the weight of the heavier particle, which is a quick sanity check on the whole solution.

Chapter summary

Idea	Key point
Force diagram	Shows every force acting on a particle: weight, normal reaction, tension, thrust, friction, driving force, resistance
Newton's first law	Zero resultant force means zero acceleration (equilibrium)
Newton's second law	$F = ma$; resultant force and acceleration are in the same direction
Newton's third law	Action–reaction pairs are equal, opposite, and act on different bodies
Connected particles	Light, inextensible string: same tension throughout, same acceleration for both particles
Pulley formulas	Two hanging masses: $a = (m_1 - m_2)g / (m_1 + m_2)$; table + hanging mass: $a = m_2g / (m_1 + m_2)$

You should now be able to

- Draw a correctly labelled force diagram for a particle, including weight, normal reaction, tension, thrust, friction, driving force and resistance.
- State and apply Newton's first and second laws, including resolving forces in one dimension and at an angle, and distinguish equilibrium from accelerated motion.
- State Newton's third law and identify action–reaction pairs correctly.

-
- Set up and solve simultaneous equations of motion for two particles connected by a light, inextensible string, on the same line, with or without a pulley.
 - Solve standard pulley problems: two particles hanging vertically, and one particle on a table connected over a pulley to a hanging particle, finding both the acceleration and the tension.